ON AN INTEGRAL SCHEME THE FIRST CHERN CLASS COINCIDES WITH THE WEIL DIVISOR ASSOCIATED TO THE INVERTIBLE SHEAF

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xypic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The defining relation of the Chern classes, obtained by expanding the cycle class of the zero scheme of a section of the tautological quotient. Same chapter chow.tex, Definition \ref{definition-chern-classes} and the equation labelled equation-chern-classes, used at proof line 7200. Specialized to the case of rank one, where $\mathbf{P}(\mathcal{L}) = X$ and $\mathcal{O}_{\mathbf{P}(\mathcal{L})}(1) = \mathcal{L}$, it reduces to $(-1)^1 c_1(\mathcal{L}) \cap c_0 + (-1)^0 c_1 = 0$, which gives the claim.
- The Weil divisor associated to an invertible sheaf. Same chapter chow.tex, Definition \ref{definition-divisor-invertible-sheaf} (invoked at proof line 7196). It supplies the construction that the proof denotes by $c_1(\mathcal{L}) \cap [X]$ and compares with the first Chern class.

Source reference. Original definition 02U0: sources/stacks-chow/pinned-original-source.tex, source lines 7144–7144. Navigation pointer for \ref{definition-chern-classes} used inside the counted span (line 7189); the argument that consumes it is retained in the span.

Source reference. Original definition 02SJ: sources/stacks-chow/pinned-original-source.tex, source lines 4149–4149. Navigation pointer for \ref{definition-divisor-invertible-sheaf} used inside the counted span (line 7191); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original equation 02U1: sources/stacks-chow/pinned-original-source.tex, source lines 7156–7156. Navigation pointer for \ref{equation-chern-classes} used inside the counted span (line 7200); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 7184); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Assume X is integral and $n = \dim_\delta(X)$. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. The first Chern class of $\mathcal{L}$ on X of Definition 02U0 is equal to the Weil divisor associated to $\mathcal{L}$ by Definition 02SJ.*

Proof. In this proof we use $c_1(\mathcal{L}) \cap [X]$ to denote the construction of Definition 02SJ. Since $\mathcal{L}$ has rank 1 we have $\mathbf{P}(\mathcal{L}) = X$ and $\mathcal{O}_{\mathbf{P}(\mathcal{L})}(1) = \mathcal{L}$ by our normalizations. Hence (02U1) reads

$$(-1)^1 c_1(\mathcal{L}) \cap c_0 + (-1)^0 c_1 = 0$$

Since $c_0 = [X]$, we conclude $c_1 = c_1(\mathcal{L}) \cap [X]$ as desired. □